

## 6. The Filtering Problem

### 6.1 Introduction

Problem 3 in the introduction is a special case of the following general *filtering problem*:

Suppose the state  $X_t \in \mathbf{R}^n$  at time  $t$  of a system is given by a stochastic differential equation

$$\frac{dX_t}{dt} = b(t, X_t) + \sigma(t, X_t)W_t, \quad t \geq 0, \quad (6.1.1)$$

where  $b: \mathbf{R}^{n+1} \rightarrow \mathbf{R}^n$ ,  $\sigma: \mathbf{R}^{n+1} \rightarrow \mathbf{R}^{n \times p}$  satisfy conditions (5.2.1), (5.2.2) and  $W_t$  is  $p$ -dimensional white noise. As discussed earlier the Itô interpretation of this equation is

$$\text{(system)} \quad dX_t = b(t, X_t)dt + \sigma(t, X_t)dU_t, \quad (6.1.2)$$

where  $U_t$  is  $p$ -dimensional Brownian motion. We also assume that the distribution of  $X_0$  is known and independent of  $U_t$ . Similarly to the 1-dimensional situation (3.3.6) there is an explicit several-dimensional formula which expresses the *Stratonovich* interpretation of (6.1.1):

$$dX_t = b(t, X_t)dt + \sigma(t, X_t) \circ dU_t$$

in terms of Itô integrals as follows:

$$\begin{aligned} dX_t &= \tilde{b}(t, X_t)dt + \sigma(t, X_t)dU_t, \quad \text{where} \\ \tilde{b}_i(t, x) &= b_i(t, x) + \frac{1}{2} \sum_{j=1}^p \sum_{k=1}^n \frac{\partial \sigma_{ij}}{\partial x_k} \sigma_{kj}; \quad 1 \leq i \leq n. \end{aligned} \quad (6.1.3)$$

(See Stratonovich (1966)). From now on we will use the Itô interpretation (6.1.2).

In the continuous version of the filtering problem we assume that the observations  $H_t \in \mathbf{R}^m$  are performed continuously and are of the form

$$H_t = c(t, X_t) + \gamma(t, X_t) \cdot \widetilde{W}_t, \quad (6.1.4)$$

where  $c: \mathbf{R}^{n+1} \rightarrow \mathbf{R}^m$ ,  $\gamma: \mathbf{R}^{n+1} \rightarrow \mathbf{R}^{m \times r}$  are functions satisfying (5.2.1) and  $\widetilde{W}_t$  denotes  $r$ -dimensional white noise, independent of  $U_t$  and  $X_0$ .

To obtain a tractable mathematical interpretation of (6.1.4) we introduce

$$Z_t = \int_0^t H_s ds \quad (6.1.5)$$

and thereby we obtain the stochastic integral representation

$$(\text{observations}) \quad dZ_t = c(t, X_t)dt + \gamma(t, X_t)dV_t, \quad Z_0 = 0 \quad (6.1.6)$$

where  $V_t$  is  $r$ -dimensional Brownian motion, independent of  $U_t$  and  $X_0$ .

Note that if  $H_s$  is known for  $0 \leq s \leq t$ , then  $Z_s$  is also known for  $0 \leq s \leq t$  and conversely. So no information is lost or gained by considering  $Z_t$  as our “observations” instead of  $H_t$ . But this allows us to obtain a well-defined mathematical model of the situation.

**The filtering problem** is the following:

Given the observations  $Z_s$  satisfying (6.1.6) for  $0 \leq s \leq t$ , what is the best estimate  $\widehat{X}_t$  of the state  $X_t$  of the system (6.1.2) based on these observations?

As we have pointed out earlier, it is necessary to find a precise mathematical formulation of this problem: By saying that the estimate  $\widehat{X}_t$  is based on the observations  $\{Z_s; s \leq t\}$  we mean that

$$\begin{aligned} \widehat{X}_t(\cdot) & \text{ is } \mathcal{G}_t\text{-measurable,} \\ \text{where } \mathcal{G}_t & \text{ is the } \sigma\text{-algebra generated by } \{Z_s(\cdot), s \leq t\}. \end{aligned} \quad (6.1.7)$$

By saying that  $\widehat{X}_t$  is the best such estimate we mean that

$$\int_{\Omega} |X_t - \widehat{X}_t|^2 dP = E[|X_t - \widehat{X}_t|^2] = \inf \{E[|X_t - Y|^2]; Y \in \mathcal{K}\}. \quad (6.1.8)$$

Here – and in the rest of this chapter –  $(\Omega, \mathcal{F}, P)$  is the probability space corresponding to the  $(p+r)$ -dimensional Brownian motion  $(U_t, V_t)$  starting at 0,  $E$  denotes expectation w.r.t.  $P$  and

$$\mathcal{K} := \mathcal{K}_t := \mathcal{K}(Z, t) := \{Y: \Omega \rightarrow \mathbf{R}^n; Y \in L^2(P) \text{ and } Y \text{ is } \mathcal{G}_t\text{-measurable}\}, \quad (6.1.9)$$

where  $L^2(P) = L^2(\Omega, P)$ .

Having found the mathematical formulation of our problem, we now start to study the properties of the solution  $\widehat{X}_t$ .

We first establish the following useful connection between conditional expectation and projection:

**Lemma 6.1.1.** *Let  $\mathcal{H} \subset \mathcal{F}$  be a  $\sigma$ -algebra and let  $X \in L^2(P)$  be  $\mathcal{F}$ -measurable. Put  $\mathcal{N} = \{Y \in L^2(P); Y \text{ is } \mathcal{H}\text{-measurable}\}$  and let  $\mathcal{P}_{\mathcal{N}}$  denote*

the (orthogonal) projection from the Hilbert space  $L^2(P)$  into the subspace  $\mathcal{N}$ . Then

$$\mathcal{P}_{\mathcal{N}}(X) = E[X|\mathcal{H}] .$$

*Proof.* Recall (see Appendix B) that  $E[X|\mathcal{H}]$  is by definition the  $P$ -unique function from  $\Omega$  to  $\mathbf{R}$  such that

- (i)  $E[X|\mathcal{H}]$  is  $\mathcal{H}$ -measurable
- (ii)  $\int_A E[X|\mathcal{H}]dP = \int_A XdP$  for all  $A \in \mathcal{H}$ .

Now  $\mathcal{P}_{\mathcal{N}}(X)$  is  $\mathcal{H}$ -measurable and

$$\int_{\Omega} Y(X - \mathcal{P}_{\mathcal{N}}(X))dP = 0 \quad \text{for all } Y \in \mathcal{N} .$$

In particular,

$$\int_A (X - \mathcal{P}_{\mathcal{N}}(X))dP = 0 \quad \text{for all } A \in \mathcal{H}$$

i.e.

$$\int_A \mathcal{P}_{\mathcal{N}}(X)dP = \int_A XdP \quad \text{for all } A \in \mathcal{H} .$$

Hence, by uniqueness,  $\mathcal{P}_{\mathcal{N}}(X) = E[X|\mathcal{H}]$ .  $\square$

From the general theory of Hilbert spaces we know that the solution  $\hat{X}_t$  of the problem (6.1.8) is given by the projection  $\mathcal{P}_{\mathcal{K}_t}(X_t)$ . Therefore Lemma 6.1.1 leads to the following useful result:

**Theorem 6.1.2.**

$$\hat{X}_t = \mathcal{P}_{\mathcal{K}_t}(X_t) = E[X_t|\mathcal{G}_t] .$$

This is the basis for the general Fujisaki-Kallianpur-Kunita equation of filtering theory. See for example Bensoussan (1992), Davis (1984) or Kallianpur (1980).

## 6.2 The 1-Dimensional Linear Filtering Problem

From now on we will concentrate on the linear case, which allows an explicit solution in terms of a stochastic differential equation for  $\hat{X}_t$  (the *Kalman-Bucy filter*):

In the *linear* filtering problem the system and observation equations have the form:

$$\begin{aligned} \text{(linear system)} \quad dX_t &= F(t)X_t dt + C(t)dU_t; \\ F(t) &\in \mathbf{R}^{n \times n}, \quad C(t) \in \mathbf{R}^{n \times p} \end{aligned} \quad (6.2.1)$$

$$\begin{aligned} \text{(linear observations)} \quad dZ_t &= G(t)X_t dt + D(t)dV_t; \\ G(t) &\in \mathbf{R}^{m \times n}, \quad D(t) \in \mathbf{R}^{m \times r} \end{aligned} \quad (6.2.2)$$

To be able to focus on the main ideas in the solution of the filtering problem, we will first consider only the 1-dimensional case:

$$\text{(linear system)} \quad dX_t = F(t)X_t dt + C(t)dU_t; \quad F(t), C(t) \in \mathbf{R} \quad (6.2.3)$$

$$\text{(linear observations)} \quad dZ_t = G(t)X_t dt + D(t)dV_t; \quad G(t), D(t) \in \mathbf{R} \quad (6.2.4)$$

We assume (see (5.2.1)) that  $F, G, C, D$  are bounded on bounded intervals. Based on our interpretation (6.1.5) of  $Z_t$  we assume  $Z_0 = 0$ . We also assume that  $X_0$  is normally distributed (and independent of  $\{U_t\}, \{V_t\}$ ). Finally we assume that  $D(t)$  is bounded away from 0 on bounded intervals.

The (important) extension to the several-dimensional case (6.2.1), (6.2.2) is technical, but does not require any essentially new ideas. Therefore we shall only state the result for this case (in the next section) after we have discussed the 1-dimensional situation. The reader is encouraged to work out the necessary modifications for the general case for himself or consult Bensoussan (1992), Davis (1977) or Kallianpur (1980) for a full treatment.

From now on we let  $X_t, Z_t$  be processes satisfying (6.2.3), (6.2.4). Here is an outline of the solution of the filtering problem in this case.

**Step 1.** Let  $\mathcal{L} = \mathcal{L}(Z, t)$  be the closure in  $L^2(P)$  of functions which are linear combinations of the form

$$c_0 + c_1 Z_{s_1}(\omega) + \cdots + c_k Z_{s_k}(\omega), \quad \text{with } s_j \leq t, c_j \in \mathbf{R}.$$

Let

$$\mathcal{P}_{\mathcal{L}} \quad \text{denote the projection from } L^2(P) \text{ onto } \mathcal{L}.$$

Then, with  $\mathcal{K}$  as in (6.1.9),

$$\hat{X}_t = \mathcal{P}_{\mathcal{K}}(X_t) = E[X_t | \mathcal{G}_t] = \mathcal{P}_{\mathcal{L}}(X_t).$$

Thus, the best  $Z$ -measurable estimate of  $X_t$  coincides with the best  $Z$ -linear estimate of  $X_t$ .

**Step 2.** Replace  $Z_t$  by the *innovation process*  $N_t$ :

$$N_t = Z_t - \int_0^t (GX)_s^\wedge ds, \quad \text{where } (GX)_s^\wedge = \mathcal{P}_{\mathcal{L}(Z, s)}(G(s)X_s) = G(s)\hat{X}_s.$$

Then

- (i)  $N_t$  has *orthogonal increments*, i.e.  
 $E[(N_{t_1} - N_{s_1})(N_{t_2} - N_{s_2})] = 0$  for non-overlapping intervals  $[s_1, t_1], [s_2, t_2]$ .

(ii)  $\mathcal{L}(N, t) = \mathcal{L}(Z, t)$ , so  $\hat{X}_t = \mathcal{P}_{\mathcal{L}(N, t)}(X_t)$ .

**Step 3.** If we put

$$dR_t = \frac{1}{D(t)} dN_t,$$

then  $R_t$  is a 1-dimensional Brownian motion. Moreover,

$$\mathcal{L}(N, t) = \mathcal{L}(R, t) \quad \text{and}$$

$$\hat{X}_t = \mathcal{P}_{\mathcal{L}(N, t)}(X_t) = \mathcal{P}_{\mathcal{L}(R, t)}(X_t) = E[X_t] + \int_0^t \frac{\partial}{\partial s} E[X_t R_s] dR_s.$$

**Step 4.** Find an expression for  $X_t$  by solving the (linear) stochastic differential equation

$$dX_t = F(t)X_t dt + C(t)dU_t.$$

**Step 5.** Substitute the formula for  $X_t$  from Step 4 into  $E[X_t R_s]$  and use Step 3 to obtain a stochastic differential equation for  $\hat{X}_t$ :

$$d\hat{X}_t = \frac{\partial}{\partial s} E[X_t R_s]_{s=t} dR_t + \left( \int_0^t \frac{\partial^2}{\partial t \partial s} E[X_t R_s] dR_s \right) dt \quad \text{etc.}$$

Before we proceed to establish Steps 1–5, let us consider a simple, but motivating example:

**Example 6.2.1.** Suppose  $X, W_1, W_2, \dots$  are independent real random variables,  $E[X] = E[W_j] = 0$  for all  $j$ ,  $E[X^2] = a^2$ ,  $E[W_j^2] = m^2$  for all  $j$ . Put  $Z_j = X + W_j$ .

What is the best *linear* estimate  $\hat{X}$  of  $X$  based on  $\{Z_j; j \leq k\}$ ? More precisely, let

$$\mathcal{L} = \mathcal{L}(Z, k) = \{c_1 Z_1 + \dots + c_k Z_k; c_1, \dots, c_k \in \mathbf{R}\}.$$

Then we want to find

$$\hat{X}_k = \mathcal{P}_k(X),$$

where  $\mathcal{P}_k$  denotes the projection into  $\mathcal{L}(Z, k)$ .

We use the Gram-Schmidt procedure to obtain random variables  $A_1, A_2, \dots$  such that

- (i)  $E[A_i A_j] = 0$  for  $i \neq j$
- (ii)  $\mathcal{L}(A, k) = \mathcal{L}(Z, k)$  for all  $k$ .

Then

$$\hat{X}_k = \sum_{j=1}^k \frac{E[XA_j]}{E[A_j^2]} A_j \quad \text{for } k = 1, 2, \dots \quad (6.2.5)$$

We obtain a recursive relation between  $\hat{X}_k$  and  $\hat{X}_{k-1}$  from this by observing that

$$A_j = Z_j - \hat{X}_{j-1} , \quad (6.2.6)$$

which follows from

$$A_j = Z_j - \mathcal{P}_{j-1}(Z_j) = Z_j - \mathcal{P}_{j-1}(X) , \quad \text{since } \mathcal{P}_{j-1}(W_j) = 0 .$$

By (6.2.6)

$$E[XA_j] = E[X(Z_j - \hat{X}_{j-1})] = E[X(X - \hat{X}_{j-1})] = E[(X - \hat{X}_{j-1})^2]$$

and

$$E[A_j^2] = E[(X + W_j - \hat{X}_{j-1})^2] = E[(X - \hat{X}_{j-1})^2] + m^2 .$$

Hence

$$\hat{X}_k = \hat{X}_{k-1} + \frac{E[(X - \hat{X}_{k-1})^2]}{E[(X - \hat{X}_{k-1})^2] + m^2} (Z_k - \hat{X}_{k-1}) . \quad (6.2.7)$$

If we introduce

$$\bar{Z}_k = \frac{1}{k} \sum_{j=1}^k Z_j ,$$

then this can be simplified to

$$\hat{X}_k = \frac{a^2}{a^2 + \frac{1}{k} m^2} \bar{Z}_k . \quad (6.2.8)$$

(This can be seen as follows:

Put

$$\alpha_k = \frac{a^2}{a^2 + \frac{1}{k} m^2} , \quad U_k = \alpha_k \bar{Z}_k .$$

Then

- (i)  $U_k \in \mathcal{L}(Z, k)$
- (ii)  $X - U_k \perp \mathcal{L}(Z, k)$ , since

$$\begin{aligned} E[(X - U_k)Z_i] &= E[XZ_i] - \alpha_k E[\bar{Z}_k Z_i] \\ &= E[X(X + W_i)] - \alpha_k \frac{1}{k} \sum_j E[Z_j Z_i] \\ &= a^2 - \frac{1}{k} \alpha_k \sum_j E[(X + W_j)(X + W_i)] = a^2 - \frac{1}{k} \alpha_k [ka^2 + m^2] = 0 . \end{aligned}$$

The result can be interpreted as follows:

For large  $k$  we put  $\hat{X}_k \approx \bar{Z}_k$ , while for small  $k$  the relation between  $a^2$  and  $m^2$  becomes more important. If  $m^2 \gg a^2$ , the observations are to a large extent neglected (for small  $k$ ) and  $\hat{X}_k$  is put equal to its mean value, 0. See also Exercise 6.11.

This example gives the motivation for our approach:

We replace the process  $Z_t$  by an orthogonal increment process  $N_t$  (Step 2) in order to obtain a representation for  $\hat{X}_t$  analogous to (6.2.5). Such a representation is obtained in Step 3, after we have identified the best *linear* estimate with the best *measurable* estimate (Step 1) and established the connection between  $N_t$  and Brownian motion.

### Step 1. Z-Linear and Z-Measurable Estimates

**Lemma 6.2.2.** *Let  $X, Z_s; s \leq t$  be random variables in  $L^2(P)$  and assume that*

$$(X, Z_{s_1}, Z_{s_2}, \dots, Z_{s_n}) \in \mathbf{R}^{n+1}$$

*has a normal distribution for all  $s_1, s_2, \dots, s_n \leq t, n \geq 1$ . Then*

$$\mathcal{P}_{\mathcal{L}}(X) = E[X|\mathcal{G}] = \mathcal{P}_{\mathcal{K}}(X) .$$

*In other words, the best Z-linear estimate for  $X$  coincides with the best Z-measurable estimate in this case.*

*Proof.* Put  $\check{X} = \mathcal{P}_{\mathcal{L}}(X)$ ,  $\tilde{X} = X - \check{X}$ . Then we claim that  $\tilde{X}$  is independent of  $\mathcal{G}$ : Recall that a random variable  $(Y_1, \dots, Y_k) \in \mathbf{R}^k$  is normal iff  $c_1 Y_1 + \dots + c_k Y_k$  is normal, for all choices of  $c_1, \dots, c_k \in \mathbf{R}$ . And an  $L^2$ -limit of normal variables is again normal (Appendix A). Therefore

$$(\tilde{X}, Z_{s_1}, \dots, Z_{s_n}) \text{ is normal for all } s_1, \dots, s_n \leq t .$$

Since  $E[\tilde{X}Z_{s_j}] = 0$ ,  $\tilde{X}$  and  $Z_{s_j}$  are uncorrelated, for  $1 \leq j \leq n$ . It follows (Appendix A) that

$$\tilde{X} \text{ and } (Z_{s_1}, \dots, Z_{s_n}) \text{ are independent .}$$

So  $\tilde{X}$  is independent from  $\mathcal{G}$  as claimed. But then

$$E[\mathcal{X}_G(X - \check{X})] = E[\mathcal{X}_G \tilde{X}] = E[\mathcal{X}_G] \cdot E[\tilde{X}] = 0 \quad \text{for all } G \in \mathcal{G}$$

i.e.  $\int_G X dP = \int_G \check{X} dP$ . Since  $\check{X}$  is  $\mathcal{G}$ -measurable, we conclude that  $\check{X} = E[X|\mathcal{G}]$ . □

There is a curious interpretation of this result: Suppose  $X, \{Z_t\}_{t \in T}$  are  $L^2(P)$ -functions with given covariances. Among all possible distributions of

$$(X, Z_{t_1}, \dots, Z_{t_n})$$

with these covariances, the *normal* distribution will be the “worst” w.r.t. estimation, in the following sense: For any distribution we have

$$E[(X - E[X|\mathcal{G}])^2] \leq E[(X - \mathcal{P}_{\mathcal{L}}(X))^2],$$

with *equality* for the normal distribution, by Lemma 6.2.2. (Note that the quantity on the right hand side only depends on the covariances, not on the distribution we might choose to obtain these covariances). For a broad discussion of similar conclusions, based on an information theoretical game between nature and the observer, see Topsøe (1978).

Finally, to be able to apply Lemma 6.2.2 to our filtering problem, we need the following result:

**Lemma 6.2.3.**

$$M_t = \begin{bmatrix} X_t \\ Z_t \end{bmatrix} \in \mathbf{R}^2 \quad \text{is a Gaussian process.}$$

*Proof.* We may regard  $M_t$  as the solution of a 2-dimensional linear stochastic differential equation of the form

$$dM_t = H(t)M_t dt + K(t)dB_t, M_0 = \begin{bmatrix} X_0 \\ 0 \end{bmatrix}; \quad (6.2.9)$$

where  $H(t) \in \mathbf{R}^{2 \times 2}$ ,  $K(t) \in \mathbf{R}^{2 \times 2}$  and  $B_t$  is 2-dimensional Brownian motion.

Use Picard iteration to solve (6.2.9), i.e. put

$$M_t^{(n+1)} = M_0 + \int_0^t H(s)M_s^{(n)} ds + \int_0^t K(s)dB_s, \quad n = 0, 1, 2, \dots \quad (6.2.10)$$

Then  $M_t^{(n)}$  is Gaussian for all  $n$  and  $M_t^{(n)} \rightarrow M_t$  in  $L^2(P)$  (see the proof of Theorem 5.2.1) and therefore  $M_t$  is Gaussian (Theorem A.7).  $\square$

## Step 2. The Innovation Process

Before we introduce the innovation process we will establish a useful representation of the functions in the space

$$\begin{aligned} \mathcal{L}(Z, T) = \text{the closure in } L^2(P) \text{ of all linear combinations} \\ c_0 + c_1 Z_{t_1} + \dots + c_k Z_{t_k}; \quad 0 \leq t_i \leq T, c_j \in \mathbf{R}. \end{aligned}$$

If  $f \in L^2[0, T]$ , note that



$$\begin{aligned}
E \left[ \left( \int_0^T f(t) dZ_t \right)^2 \right] &= E \left[ \left( \int_0^T f(t) G(t) X_t dt \right)^2 \right] + E \left[ \left( \int_0^T f(t) D(t) dV_t \right)^2 \right] \\
&\quad + 2E \left[ \left( \int_0^T f(t) G(t) X_t dt \right) \left( \int_0^T f(t) D(t) dV_t \right) \right].
\end{aligned}$$

Since

$$\begin{aligned}
E \left[ \left( \int_0^T f(t) G(t) X_t dt \right)^2 \right] &\leq A_1 \cdot \int_0^T f(t)^2 dt \text{ by the Cauchy-Schwartz inequality,} \\
E \left[ \left( \int_0^T f(t) D(t) dV_t \right)^2 \right] &= \int_0^T f(t)^2 D^2(t) dt \text{ by the Itô isometry}
\end{aligned}$$

and  $\{X_t\}, \{V_t\}$  are independent, we conclude that

$$A_0 \int_0^T f^2(t) dt \leq E \left[ \left( \int_0^T f(t) dZ_t \right)^2 \right] \leq A_2 \int_0^T f^2(t) dt, \quad (6.2.11)$$

for some constants  $A_0, A_1, A_2$  not depending on  $f$ . We can now show

**Lemma 6.2.4.**  $\mathcal{L}(Z, T) = \{c_0 + \int_0^T f(t) dZ_t; f \in L^2[0, T], c_0 \in \mathbf{R}\}.$

*Proof.* Denote the right hand side by  $\mathcal{N}(Z, T)$ . It is enough to show that

- a)  $\mathcal{N}(Z, T) \subset \mathcal{L}(Z, T)$
- b)  $\mathcal{N}(Z, T)$  contains all linear combinations of the form

$$c_0 + c_1 Z_{t_1} + \cdots + c_k Z_{t_k}; \quad 0 \leq t_i \leq T$$

- c)  $\mathcal{N}(Z, T)$  is closed in  $L^2(P)$

a): This follows from the fact that if  $f$  is continuous then

$$\int_0^T f(t) dZ_t = \lim_{n \rightarrow \infty} \sum_j f(j \cdot 2^{-n}) \cdot (Z_{(j+1)2^{-n}} - Z_{j \cdot 2^{-n}}).$$

b): Suppose  $0 \leq t_1 < t_2 < \cdots < t_k \leq T$ . We can write

$$\sum_{i=1}^k c_i Z_{t_i} = \sum_{j=0}^{k-1} c'_j \Delta Z_j = \sum_{j=0}^{k-1} \int_{t_j}^{t_{j+1}} c'_j dZ_t = \int_0^T \left( \sum_{j=0}^{k-1} c'_j \chi_{[t_j, t_{j+1})}(t) \right) dZ_t,$$

where  $\Delta Z_j = Z_{t_{j+1}} - Z_{t_j}$ .

c): This follows from (6.2.11) and the fact that  $L^2[0, T]$  is complete.  $\square$

Now we define the **innovation process**  $N_t$  as follows:

$$N_t = Z_t - \int_0^t (GX)_s^\wedge ds, \text{ where } (GX)_s^\wedge = \mathcal{P}_{\mathcal{L}(Z,s)}(G(s)X_s) = G(s)\widehat{X}_s . \quad (6.2.12)$$

i.e.

$$dN_t = G(t)(X_t - \widehat{X}_t)dt + D(t)dV_t . \quad (6.2.13)$$

**Lemma 6.2.5.** (i)  $N_t$  has orthogonal increments

$$(ii) \quad E[N_t^2] = \int_0^t D^2(s)ds$$

$$(iii) \quad \mathcal{L}(N, t) = \mathcal{L}(Z, t) \text{ for all } t \geq 0$$

$$(iv) \quad N_t \text{ is a Gaussian process}$$

*Proof.* (i): If  $s < t$  and  $Y \in \mathcal{L}(Z, s)$  we have

$$\begin{aligned} E[(N_t - N_s)Y] &= E\left[\left(\int_s^t G(r)(X_r - \widehat{X}_r)dr + \int_s^t D(r)dV_r\right)Y\right] \\ &= \int_s^t G(r)E[(X_r - \widehat{X}_r)Y]dr + E\left[\left(\int_s^t DdV\right)Y\right] = 0 , \end{aligned}$$

since  $X_r - \widehat{X}_r \perp \mathcal{L}(Z, r) \supset \mathcal{L}(Z, s)$  for  $r \geq s$  and  $V$  has independent increments.

(ii): By Itô's formula, with  $g(t, x) = x^2$ , we have

$$d(N_t^2) = 2N_t dN_t + \frac{1}{2}2(dN_t)^2 = 2N_t dN_t + D^2 dt .$$

So

$$E[N_t^2] = E\left[\int_0^t 2N_s dN_s\right] + \int_0^t D^2(s)ds .$$

Now

$$\int_0^t N_s dN_s = \lim_{\Delta t_j \rightarrow 0} \sum N_{t_j} [N_{t_{j+1}} - N_{t_j}] ,$$

so since  $N$  has orthogonal increments we have

$$E\left[\int_0^t N_s dN_s\right] = 0 , \quad \text{and (ii) follows .}$$

(iii): It is clear that  $\mathcal{L}(N, t) \subset \mathcal{L}(Z, t)$  for all  $t \geq 0$ . To establish the opposite inclusion we use Lemma 6.2.4. So choose  $f \in L^2[0, t]$  and let us see what functions can be obtained in the form

$$\begin{aligned}
\int_0^t f(s) dN_s &= \int_0^t f(s) dZ_s - \int_0^t f(r) G(r) \widehat{X}_r dr \\
&= \int_0^t f(s) dZ_s - \int_0^t f(r) \left[ \int_0^r g(r, s) dZ_s \right] dr - \int_0^t f(r) c(r) dr \\
&= \int_0^t \left[ f(s) - \int_s^t f(r) g(r, s) dr \right] dZ_s - \int_0^t f(r) c(r) dr,
\end{aligned}$$

where we have used Lemma 6.2.2 and Lemma 6.2.4 to write, for each  $r$ ,

$$(GX)_r^\wedge = c(r) + \int_0^r g(r, s) dZ_s \quad \text{for some } g(r, \cdot) \in L^2[0, r], c(r) \in \mathbf{R}.$$

From the theory of Volterra integral equations (see e.g. Davis (1977), p. 125) there exists for all  $h \in L^2[0, t]$  an  $f \in L^2[0, t]$  such that

$$f(s) - \int_s^t f(r) g(r, s) dr = h(s).$$

So by choosing  $h = \mathcal{X}_{[0, t_1]}$  where  $0 \leq t_1 \leq t$ , we obtain

$$\int_0^t f(r) c(r) dr + \int_0^t f(s) dN_s = \int_0^t \mathcal{X}_{[0, t_1]}(s) dZ_s = Z_{t_1},$$

which shows that  $\mathcal{L}(N, t) \supset \mathcal{L}(Z, t)$ .

(iv):  $\widehat{X}_t$  is a limit (in  $L^2(P)$ ) of linear combinations of the form

$$M = c_0 + c_1 Z_{s_1} + \cdots + c_k Z_{s_k}, \quad \text{where } s_k \leq t.$$

Therefore

$$(\widehat{X}_{t_1}, \dots, \widehat{X}_{t_m})$$

is a limit of  $m$ -dimensional random variables  $(M^{(1)}, \dots, M^{(m)})$  whose components  $M^{(j)}$  are linear combinations of this form.  $(M^{(1)}, \dots, M^{(m)})$  has a normal distribution since  $\{Z_t\}$  is Gaussian, and therefore the limit has. Hence  $\{\widehat{X}_t\}$  is Gaussian. It follows that

$$N_t = Z_t - \int_0^t G(s) \widehat{X}_s ds$$

is Gaussian, by a similar argument.  $\square$

**Step 3. The Innovation Process and Brownian Motion**

Let  $N_t = Z_t - \int_0^t G(s) \widehat{X}_s ds$  be the innovation process defined in Step 2.

Recall that we have assumed that  $D(t)$  is bounded away from 0 on bounded intervals. Define the process  $R_t(\omega)$  by

$$dR_t = \frac{1}{D(t)} dN_t(\omega) ; \quad t \geq 0, \quad R_0 = 0 . \quad (6.2.14)$$

**Lemma 6.2.6.**  $R_t$  is a 1-dimensional Brownian motion.

*Proof.* Observe that

- (i)  $R_t$  has continuous paths
- (ii)  $R_t$  has orthogonal increments (since  $N_t$  has)
- (iii)  $R_t$  is Gaussian (since  $N_t$  is)
- (iv)  $E[R_t] = 0$  and  $E[R_t R_s] = \min(s, t)$ .

To prove the last assertion in (iv), note that by Itô's formula

$$d(R_t^2) = 2R_t dR_t + (dR_t)^2 = 2R_t dR_t + dt ,$$

so, since  $R_t$  has orthogonal increments,

$$E[R_t^2] = E\left[\int_0^t ds\right] = t .$$

Therefore, if  $s < t$ ,

$$E[R_t R_s] = E[(R_t - R_s) R_s] + E[R_s^2] = E[R_s^2] = s .$$

Properties (i), (iii) and (iv) constitute one of the many characterizations of a 1-dimensional Brownian motion (see Simon (1979), Theorem 4.3). (Alternatively, we could easily deduce that  $R_t$  has stationary, independent increments and therefore – by continuity – must be Brownian motion, by the result previously referred to in the beginning of Chapter 3. For a general characterization of Brownian motion see Corollary 8.4.5.)  $\square$

Since

$$\mathcal{L}(N, t) = \mathcal{L}(R, t)$$

we conclude that

$$\widehat{X}_t = \mathcal{P}_{\mathcal{L}(R, t)}(X_t) .$$

It turns out that the projection down to the space  $\mathcal{L}(R, t)$  can be described very nicely: (compare with formula (6.2.5) in Example 6.2.1)

**Lemma 6.2.7.**

$$\widehat{X}_t = E[X_t] + \int_0^t \frac{\partial}{\partial s} E[X_t R_s] dR_s . \quad (6.2.15)$$

*Proof.* From Lemma 6.2.4 we know that

$$\widehat{X}_t = c_0(t) + \int_0^t g(s) dR_s \quad \text{for some } g \in L^2[0, t], \quad c_0(t) \in \mathbf{R} .$$

Taking expectations we see that  $c_0(t) = E[\widehat{X}_t] = E[X_t]$ . We have

$$(X_t - \widehat{X}_t) \perp \int_0^t f(s) dR_s \quad \text{for all } f \in L^2[0, t] .$$

Therefore

$$\begin{aligned} E \left[ X_t \int_0^t f(s) dR_s \right] &= E \left[ \widehat{X}_t \int_0^t f(s) dR_s \right] = E \left[ \int_0^t g(s) dR_s \int_0^t f(s) dR_s \right] \\ &= E \left[ \int_0^t g(s) f(s) ds \right] = \int_0^t g(s) f(s) ds , \quad \text{for all } f \in L^2[0, t] , \end{aligned}$$

where we have used the Itô isometry. In particular, if we choose  $f = \mathcal{X}_{[0, r]}$  for some  $r \leq t$ , we obtain

$$E[X_t R_r] = \int_0^r g(s) ds$$

or

$$g(r) = \frac{\partial}{\partial r} E[X_t R_r] , \quad \text{as asserted .}$$

This completes Step 3. □

#### Step 4. An Explicit Formula for $X_t$

This is easily obtained using Itô's formula, as in the examples in Chapter 5. The result is

$$\begin{aligned} X_t &= \exp \left( \int_0^t F(s) ds \right) \left[ X_0 + \int_0^t \exp \left( - \int_0^s F(u) du \right) C(s) dU_s \right] \\ &= \exp \left( \int_0^t F(s) ds \right) X_0 + \int_0^t \exp \left( \int_s^t F(u) du \right) C(s) dU_s . \end{aligned}$$

In particular, we note that  $E[X_t] = E[X_0] \exp\left(\int_0^t F(s)ds\right)$ .

More generally, if  $0 \leq r \leq t$ , (see Exercise 6.12)

$$X_t = \exp\left(\int_r^t F(s)ds\right)X_r + \int_r^t \exp\left(\int_s^t F(u)du\right)C(s)dU_s. \quad (6.2.16)$$

### Step 5. The Stochastic Differential Equation for $\widehat{X}_t$

We now combine the previous steps to obtain the solution of the filtering problem, i.e. a stochastic differential equation for  $\widehat{X}_t$ . Starting with the formula from Lemma 6.2.7

$$\widehat{X}_t = E[X_t] + \int_0^t f(s, t) dR_s,$$

where

$$f(s, t) = \frac{\partial}{\partial s} E[X_t R_s], \quad (6.2.17)$$

we use that

$$R_s = \int_0^s \frac{G(r)}{D(r)} (X_r - \widehat{X}_r) dr + V_s \quad \text{from (6.2.13) and (6.2.14)}$$

and obtain

$$E[X_t R_s] = \int_0^s \frac{G(r)}{D(r)} E[X_t \widetilde{X}_r] dr,$$

where

$$\widetilde{X}_r = X_r - \widehat{X}_r. \quad (6.2.18)$$

Using formula (6.2.16) for  $X_t$ , we obtain

$$E[X_t \widetilde{X}_r] = \exp\left(\int_r^t F(v)dv\right) E[X_r \widetilde{X}_r] = \exp\left(\int_r^t F(v)dv\right) S(r),$$

where

$$S(r) = E[(\widetilde{X}_r)^2], \quad (6.2.19)$$

i.e. the mean square error of the estimate at time  $r \geq 0$ . Thus

$$E[X_t R_s] = \int_0^s \frac{G(r)}{D(r)} \exp\left(\int_r^t F(v)dv\right) S(r) dr$$

so that

$$f(s, t) = \frac{G(s)}{D(s)} \exp \left( \int_s^t F(v) dv \right) S(s) . \quad (6.2.20)$$

We claim that  $S(t)$  satisfies the (deterministic) differential equation

$$\frac{dS}{dt} = 2F(t)S(t) - \frac{G^2(t)}{D^2(t)}S^2(t) + C^2(t) \quad (\text{The Riccati equation}) . \quad (6.2.21)$$

To prove (6.2.21) note that by the Pythagorean theorem, (6.2.15) and the Itô isometry

$$\begin{aligned} S(t) &= E[(X_t - \hat{X}_t)^2] = E[X_t^2] - 2E[X_t \hat{X}_t] + E[\hat{X}_t^2] = E[X_t^2] - E[\hat{X}_t^2] \\ &= T(t) - \int_0^t f(s, t)^2 ds - E[X_t]^2 , \end{aligned} \quad (6.2.22)$$

where

$$T(t) = E[X_t^2] . \quad (6.2.23)$$

Now by (6.2.16) and the Itô isometry we have

$$T(t) = \exp \left( 2 \int_0^t F(s) ds \right) E[X_0^2] + \int_0^t \exp \left( 2 \int_s^t F(u) du \right) C^2(s) ds ,$$

using that  $X_0$  is independent of  $\{U_s\}_{s \geq 0}$ . So

$$\begin{aligned} \frac{dT}{dt} &= 2F(t) \cdot \exp \left( 2 \int_0^t F(s) ds \right) E[X_0^2] + C^2(t) \\ &\quad + \int_0^t 2F(t) \exp \left( 2 \int_s^t F(u) du \right) C^2(s) ds \end{aligned}$$

i.e.

$$\frac{dT}{dt} = 2F(t)T(t) + C^2(t) . \quad (6.2.24)$$

Substituting in (6.2.22) we obtain, using Step 4,

$$\begin{aligned} \frac{dS}{dt} &= \frac{dT}{dt} - f(t, t)^2 - \int_0^t 2f(s, t) \cdot \frac{\partial}{\partial t} f(s, t) ds - 2F(t)E[X_t]^2 \\ &= 2F(t)T(t) + C^2(t) - \frac{G^2(t)S^2(t)}{D^2(t)} - 2 \int_0^t f^2(s, t)F(t)ds - 2F(t)E[X_t]^2 \\ &= 2F(t)S(t) + C^2(t) - \frac{G^2(t)S^2(t)}{D^2(t)} , \quad \text{which is (6.2.21)} . \end{aligned}$$

We are now ready for the stochastic differential equation for  $\hat{X}_t$ :  
From the formula

$$\hat{X}_t = c_0(t) + \int_0^t f(s, t) dR_s \quad \text{where } c_0(t) = E[X_t]$$

it follows that

$$d\hat{X}_t = c'_0(t)dt + f(t, t)dR_t + \left( \int_0^t \frac{\partial}{\partial t} f(s, t) dR_s \right) dt, \quad (6.2.25)$$

since

$$\begin{aligned} \int_0^u \left( \int_0^t \frac{\partial}{\partial t} f(s, t) dR_s \right) dt &= \int_0^u \left( \int_s^u \frac{\partial}{\partial t} f(s, t) dt \right) dR_s \\ &= \int_0^u (f(s, u) - f(s, s)) dR_s = \hat{X}_u - c_0(u) - \int_0^u f(s, s) dR_s. \end{aligned}$$

So

$$d\hat{X}_t = c'_0(t)dt + \frac{G(t)S(t)}{D(t)}dR_t + \left( \int_0^t f(s, t) dR_s \right) F(t)dt$$

or

$$\begin{aligned} d\hat{X}_t &= c'_0(t)dt + F(t) \cdot (\hat{X}_t - c_0(t))dt + \frac{G(t)S(t)}{D(t)}dR_t \\ &= F(t)\hat{X}_t dt + \frac{G(t)S(t)}{D(t)}dR_t, \end{aligned} \quad (6.2.26)$$

since  $c'_0(t) = F(t)c_0(t)$  (Step 4).

If we substitute

$$dR_t = \frac{1}{D(t)}[dZ_t - G(t)\hat{X}_t dt]$$

we obtain

$$d\hat{X}_t = \left( F(t) - \frac{G^2(t)S(t)}{D^2(t)} \right) \hat{X}_t dt + \frac{G(t)S(t)}{D^2(t)} dZ_t. \quad (6.2.27)$$

So the conclusion is:

**Theorem 6.2.8 (The 1-dimensional Kalman-Bucy filter).**

*The solution  $\hat{X}_t = E[X_t | \mathcal{G}_t]$  of the 1-dimensional linear filtering problem*

$$\text{(linear system)} \quad dX_t = F(t)X_t dt + C(t)dU_t; \quad F(t), C(t) \in \mathbf{R} \quad (6.2.3)$$

$$\text{(linear observations)} \quad dZ_t = G(t)X_t dt + D(t)dV_t; \quad G(t), D(t) \in \mathbf{R} \quad (6.2.4)$$



(with conditions as stated earlier) satisfies the stochastic differential equation

$$d\hat{X}_t = \left( F(t) - \frac{G^2(t)S(t)}{D^2(t)} \right) \hat{X}_t dt + \frac{G(t)S(t)}{D^2(t)} dZ_t ; \quad \hat{X}_0 = E[X_0] \quad (6.2.28)$$

where

$$S(t) = E[(X_t - \hat{X}_t)^2] \text{ satisfies the (deterministic) Riccati equation} \\ \frac{dS}{dt} = 2F(t)S(t) - \frac{G^2(t)}{D^2(t)} S^2(t) + C^2(t), \quad S(0) = E[(X_0 - E[X_0])^2]. \quad (6.2.29)$$

**Example 6.2.9 (Noisy observations of a constant process).**

Consider the simple case

$$\begin{aligned} \text{(system)} \quad & dX_t = 0, \text{ i.e. } X_t = X_0; \quad E[X_0] = 0, \quad E[X_0^2] = a^2 \\ \text{(observations)} \quad & dZ_t = X_t dt + m dV_t; \quad Z_0 = 0 \end{aligned}$$

(corresponding to

$$H_t = \frac{dZ}{dt} = X_t + m W_t, \quad W_t = \text{white noise}).$$

First we solve the corresponding Riccati equation for

$$\begin{aligned} S(t) &= E[(X_t - \hat{X}_t)^2]: \\ \frac{dS}{dt} &= -\frac{1}{m^2} S^2, \quad S(0) = a^2 \end{aligned}$$

i.e.

$$S(t) = \frac{a^2 m^2}{m^2 + a^2 t}; \quad t \geq 0.$$

This gives the following equation for  $\hat{X}_t$ :

$$d\hat{X}_t = -\frac{a^2}{m^2 + a^2 t} \hat{X}_t dt + \frac{a^2}{m^2 + a^2 t} dZ_t; \quad \hat{X}_0 = E[X_0] = 0$$

or

$$d\left( \hat{X}_t \exp\left( \int \frac{a^2}{m^2 + a^2 t} dt \right) \right) = \exp\left( \int \frac{a^2}{m^2 + a^2 t} dt \right) \frac{a^2}{m^2 + a^2 t} dZ_t$$

which gives

$$\hat{X}_t = \frac{a^2}{m^2 + a^2 t} Z_t; \quad t \geq 0. \quad (6.2.30)$$

This is the continuous analogue of Example 6.2.1.

**Example 6.2.10 (Noisy observations of a Brownian motion).**

If we modify the preceding example slightly, so that

$$\begin{aligned} \text{(system)} \quad & dX_t = cdU_t; \quad E[X_0] = 0, \quad E[X_0^2] = a^2, \quad c \text{ constant} \\ \text{(observations)} \quad & dZ_t = X_t dt + mdV_t, \end{aligned}$$

the Riccati equation becomes

$$\frac{dS}{dt} = -\frac{1}{m^2}S^2 + c^2, \quad S(0) = a^2$$

or

$$\frac{m^2 dS}{m^2 c^2 - S^2} = dt, \quad (S \neq mc).$$

This gives

$$\left| \frac{mc + s}{mc - s} \right| = K \exp\left(\frac{2ct}{m}\right); \quad K = \left| \frac{mc + a^2}{mc - a^2} \right|.$$

Or

$$S(t) = \begin{cases} mc \frac{K \cdot \exp(\frac{2ct}{m}) - 1}{K \cdot \exp(\frac{2ct}{m}) + 1}; & \text{if } S(0) < mc \\ mc \text{ (constant)} & \text{if } S(0) = mc \\ mc \frac{K \cdot \exp(\frac{2ct}{m}) + 1}{K \cdot \exp(\frac{2ct}{m}) - 1} & \text{if } S(0) > mc. \end{cases}$$

Thus in all cases the mean square error  $S(t)$  tends to  $mc$  as  $t \rightarrow \infty$ .

For simplicity let us put  $a = 0$ ,  $m = c = 1$ . Then

$$S(t) = \frac{\exp(2t) - 1}{\exp(2t) + 1} = \tanh(t).$$

The equation for  $\hat{X}_t$  is

$$d\hat{X}_t = -\tanh(t)\hat{X}_t dt + \tanh(t)dZ_t, \quad \hat{X}_0 = 0$$

or

$$d(\cosh(t)\hat{X}_t) = \sinh(t)dZ_t.$$

So

$$\hat{X}_t = \frac{1}{\cosh(t)} \int_0^t \sinh(s) dZ_s.$$

If we return to the interpretation of  $Z_t$ :

$$Z_t = \int_0^t H_s ds,$$

where  $H_s$  are the “original” observations (see (6.1.4)), we can write

$$\hat{X}_t = \frac{1}{\cosh(t)} \int_0^t \sinh(s) H_s ds, \quad (6.2.31)$$

so  $\hat{X}_t$  is approximately (for large  $t$ ) a weighted average of the observations  $H_s$ , with increasing emphasis on observations as time increases.

**Remark.** It is interesting to compare formula (6.2.31) with established formulas in forecasting. For example, the *exponentially weighted moving average*  $\tilde{X}_n$  suggested by C.C. Holt in 1958 is given by

$$\tilde{X}_n = (1 - \alpha)^n Z_0 + \alpha \sum_{k=1}^n (1 - \alpha)^{n-k} Z_k,$$

where  $\alpha$  is some constant;  $0 \leq \alpha \leq 1$ . See The Open University (1981), p. 16.

This may be written

$$\tilde{X}_n = \beta^{-n} Z_0 + (\beta - 1) \beta^{-n-1} \sum_{k=1}^n \beta^k Z_k,$$

where  $\beta = \frac{1}{1-\alpha}$  (assuming  $\alpha < 1$ ), which is a discrete version of (6.2.31), or – more precisely – of the formula corresponding to (6.2.31) in the general case when  $a \neq 0$  and  $m, c$  are not necessarily equal to 1.

**Example 6.2.11 (Estimation of a parameter).**

Suppose we want to estimate the value of a (constant) parameter  $\theta$ , based on observations  $Z_t$  satisfying the model

$$dZ_t = \theta M(t)dt + N(t)dB_t,$$

where  $M(t), N(t)$  are known functions. In this case the stochastic differential equation for  $\theta$  is of course

$$d\theta = 0,$$

so the Riccati equation for  $S(t) = E[(\theta - \hat{\theta}_t)^2]$  is

$$\frac{dS}{dt} = - \left( \frac{M(t)S(t)}{N(t)} \right)^2$$

which gives

$$S(t) = \left( S_0^{-1} + \int_0^t M(s)^2 N(s)^{-2} ds \right)^{-1}$$

and the Kalman-Bucy filter is

$$d\hat{\theta}_t = \frac{M(t)S(t)}{N(t)^2} (dZ_t - M(t)\hat{\theta}_t dt).$$

This can be written

$$\left( S_0^{-1} + \int_0^t M(s)^2 N(s)^{-2} ds \right) d\hat{\theta}_t + M(t)^2 N(t)^{-2} \hat{\theta}_t dt = M(t) N(t)^{-2} dZ_t .$$

We recognize the left hand side as

$$d\left( \left( S_0^{-1} + \int_0^t M(s)^2 N(s)^{-2} ds \right) \hat{\theta}_t \right)$$

so we obtain

$$\hat{\theta}_t = \frac{\hat{\theta}_0 S_0^{-1} + \int_0^t M(s) N(s)^{-2} dZ_s}{S_0^{-1} + \int_0^t M(s)^2 N(s)^{-2} ds} .$$

This estimate coincides with the maximum likelihood estimate in classical estimation theory if  $S_0^{-1} = 0$ . See Liptser and Shiryaev (1978).

For more information about estimates of drift parameters in diffusions and generalizations, see for example Aase (1982), Brown and Hewitt (1975) and Taraskin (1974).

**Example 6.2.12 (Noisy observations of a population growth).**

Consider a simple growth model ( $r$  constant)

$$dX_t = rX_t dt, E[X_0] = b > 0, \quad E[(X_0 - b)^2] = a^2 ,$$

with observations

$$dZ_t = X_t dt + m dV_t ; \quad m \text{ constant} .$$

The corresponding Riccati equation

$$\frac{dS}{dt} = 2rS - \frac{1}{m^2} S^2, \quad S(0) = a^2 ,$$

gives the logistic curve

$$S(t) = \frac{2rm^2}{1 + Ke^{-2rt}} ; \quad \text{where } K = \frac{2rm^2}{a^2} - 1 .$$

So the equation for  $\hat{X}_t$  becomes

$$d\hat{X}_t = \left( r - \frac{S}{m^2} \right) \hat{X}_t dt + \frac{S}{m^2} dZ_t ; \quad \hat{X}_0 = E[X_0] = b .$$

For simplicity let us assume that  $a^2 = 2rm^2$ , so that

$$S(t) = 2rm^2 \quad \text{for all } t.$$

(In the general case  $S(t) \rightarrow 2rm^2$  as  $t \rightarrow \infty$ , so this is not an unreasonable approximation for large  $t$ ). Then we get

$$d(\exp(rt)\hat{X}_t) = \exp(rt)2rdZ_t, \quad \hat{X}_0 = b$$

or

$$\hat{X}_t = \exp(-rt) \left[ \int_0^t 2r \exp(rs) dZ_s + b \right].$$

As in Example 6.2.10 this may be written

$$\hat{X}_t = \exp(-rt) \left[ \int_0^t 2r \exp(rs) H_s ds + b \right], \quad \text{if } Z_t = \int_0^t H_s ds. \quad (6.2.32)$$

For example, assume that  $H_s = \beta$  (constant) for  $0 \leq s \leq t$ , i.e. that our observations (for some reason) give the same value  $\beta$  for all times  $s \leq t$ . Then

$$\hat{X}_t = 2\beta - (2\beta - b) \exp(-rt) \rightarrow 2\beta \quad \text{as } t \rightarrow \infty.$$

If  $H_s = \beta \cdot \exp(\alpha s)$ ,  $s \geq 0$  ( $\alpha$  constant), we get

$$\begin{aligned} \hat{X}_t &= \exp(-rt) \left[ \frac{2r\beta}{r+\alpha} (\exp(r+\alpha)t - 1) + b \right] \\ &\approx \frac{2r\beta}{r+\alpha} \exp \alpha t \quad \text{for large } t. \end{aligned}$$

Thus, only if  $\alpha = r$ , i.e.  $H_s = \beta \exp(rs)$ ;  $s \geq 0$ , does the filter “believe” the observations in the long run. And only if  $\alpha = r$  and  $\beta = b$ , i.e.  $H_s = b \exp(rs)$ ;  $s \geq 0$ , does the filter “believe” the observations at all times.

**Example 6.2.13 (Constant coefficients – general discussion).**

Now consider the system

$$dX_t = FX_t dt + CdU_t; \quad F, C \text{ constants } \neq 0$$

with observations

$$dZ_t = GX_t dt + DdV_t; \quad G, D \text{ constants } \neq 0.$$

The corresponding Riccati equation

$$S' = 2FS - \frac{G^2}{D^2} S^2 + C^2, \quad S(0) = a^2$$

has the solution

$$S(t) = \frac{\alpha_1 - K\alpha_2 \exp(\frac{(\alpha_2 - \alpha_1)G^2 t}{D^2})}{1 - K \exp(\frac{(\alpha_2 - \alpha_1)G^2 t}{D^2})},$$

where

$$\begin{aligned}\alpha_1 &= G^{-2}(FD^2 - D\sqrt{F^2 D^2 + G^2 C^2}) \\ \alpha_2 &= G^{-2}(FD^2 + D\sqrt{F^2 D^2 + G^2 C^2})\end{aligned}$$

and

$$K = \frac{a^2 - \alpha_1}{a^2 - \alpha_2}.$$

This gives the solution for  $\hat{X}_t$  of the form

$$\hat{X}_t = \exp\left(\int_0^t H(s)ds\right)\hat{X}_0 + \frac{G}{D^2} \int_0^t \exp\left(\int_s^t H(u)du\right)S(s)dZ_s,$$

where

$$H(s) = F - \frac{G^2}{D^2}S(s).$$

For large  $s$  we have  $S(s) \approx \alpha_2$ . This gives

$$\begin{aligned}\hat{X}_t &\approx \hat{X}_0 \exp\left(\left(F - \frac{G^2 \alpha_2}{D^2}\right)t\right) + \frac{G\alpha_2}{D^2} \int_0^t \exp\left(\left(F - \frac{G^2 \alpha_2}{D^2}\right)(t-s)\right)dZ_s \\ &= \hat{X}_0 \exp(-\beta t) + \frac{G\alpha_2}{D^2} \exp(-\beta t) \int_0^t \exp(\beta s)dZ_s\end{aligned}\quad (6.2.33)$$

where  $\beta = D^{-1}\sqrt{F^2 D^2 + G^2 C^2}$ . So we get approximately the same behaviour as in the previous example.

### 6.3 The Multidimensional Linear Filtering Problem

Finally we formulate the solution of the  $n$ -dimensional linear filtering problem (6.2.1), (6.2.2):

**Theorem 6.3.1 (The Multi-Dimensional Kalman-Bucy Filter).**

*The solution  $\hat{X}_t = E[X_t|\mathcal{G}_t]$  of the multi-dimensional linear filtering problem*

$$\begin{aligned}(\text{linear system}) \quad dX_t &= F(t)X_t dt + C(t)dU_t; \\ &F(t) \in \mathbf{R}^{n \times n}, \quad C(t) \in \mathbf{R}^{n \times p}\end{aligned}\quad (6.3.1)$$

$$\begin{aligned}(\text{linear observations}) \quad dZ_t &= G(t)X_t dt + D(t)dV_t; \\ &G(t) \in \mathbf{R}^{m \times n}, \quad D(t) \in \mathbf{R}^{m \times r}\end{aligned}\quad (6.3.2)$$

satisfies the stochastic differential equation

$$d\hat{X}_t = (F - SG^T(DD^T)^{-1}G)\hat{X}_t dt + SG^T(DD^T)^{-1}dZ_t ; \quad \hat{X}_0 = E[X_0] \quad (6.3.3)$$

where  $S(t) := E[(X_t - \hat{X}_t)(X_t - \hat{X}_t)^T] \in \mathbf{R}^{n \times n}$  satisfies the matrix Riccati equation

$$\begin{aligned} \frac{dS}{dt} &= FS + SF^T - SG^T(DD^T)^{-1}GS + CC^T ; \\ S(0) &= E[(X_0 - E[X_0])(X_0 - E[X_0])^T] . \end{aligned} \quad (6.3.4)$$

The condition on  $D(t) \in \mathbf{R}^{m \times r}$  is now that  $D(t)D(t)^T$  is invertible for all  $t$  and that  $(D(t)D(t)^T)^{-1}$  is bounded on every bounded  $t$ -interval.

A similar solution can be found for the more general situation

$$\text{(system)} \quad dX_t = [F_0(t) + F_1(t)X_t + F_2(t)Z_t]dt + C(t)dU_t \quad (6.3.5)$$

$$\text{(observations)} \quad dZ_t = [G_0(t) + G_1(t)X_t + G_2(t)Z_t]dt + D(t)dV_t , \quad (6.3.6)$$

where  $X_t \in \mathbf{R}^n$ ,  $Z_t \in \mathbf{R}^m$  and  $B_t = (U_t, V_t)$  is  $n + m$ -dimensional Brownian motion, with appropriate dimensions on the matrix coefficients. See Bensoussan (1992) and Kallianpur (1980), who also treat the non-linear case. An account of non-linear filtering theory is also given in Pardoux (1979) and Davis (1984).

For the solution of linear filtering problems governed by more general processes than Brownian motion (processes with orthogonal increments) see Davis (1977).

For various applications of filtering theory see Bucy and Joseph (1968), Jazwinski (1970), Gelb (1974), Maybeck (1979) and the references in these books.

## Exercises

### 6.1. (Time-varying observations of a constant)

Prove that if the (1-dimensional) system is

$$dX_t = 0, \quad E[X_0] = 0, \quad E[X_0^2] = a^2$$

and the observation process is

$$dZ_t = G(t)X_t dt + dV_t, \quad Z_0 = 0$$

then  $S(t) = E[(X_t - \hat{X}_t)^2]$  is given by

$$S(t) = \frac{1}{\frac{1}{S(0)} + \int_0^t G^2(s)ds} . \quad (6.3.7)$$

We say that we have *exact asymptotic estimation* if  $S(t) \rightarrow 0$  as  $t \rightarrow \infty$ , i.e. if

$$\int_0^\infty G^2(s)ds = \infty .$$

Thus for

$$G(s) = \frac{1}{(1+s)^p} \quad (p > 0 \text{ constant})$$

we have exact asymptotic estimation iff  $p \leq \frac{1}{2}$ .

**6.2.** Consider the linear 1-dimensional filtering problem with no noise in the system:

$$\text{(system)} \quad dX_t = F(t)X_t dt \quad (6.3.8)$$

$$\text{(observations)} \quad dZ_t = G(t)X_t dt + D(t)dV_t \quad (6.3.9)$$

Put  $S(t) = E[(X_t - \hat{X}_t)^2]$  as usual and assume  $S(0) > 0$ .

a) Show that

$$R(t) := \frac{1}{S(t)}$$

satisfies the *linear* differential equation

$$R'(t) = -2F(t)R(t) + \frac{G^2(t)}{D^2(t)}; \quad R(0) = \frac{1}{S(0)} \quad (6.3.10)$$

b) Use (6.3.10) to prove that for the filtering problem (6.3.8), (6.3.9) we have

$$\frac{1}{S(t)} = \frac{1}{S(0)} \exp \left( -2 \int_0^t F(s)ds \right) + \int_0^t \exp \left( -2 \int_s^t F(u)du \right) \frac{G^2(s)}{D^2(s)} ds . \quad (6.3.11)$$

**6.3.** In Example 6.2.12 we found that

$$S(t) \rightarrow 2rm^2 \quad \text{as } t \rightarrow \infty ,$$

so exact asymptotic estimation (Exercise 6.1) of  $X_t$  is not possible. However, prove that we can obtain exact asymptotic estimation of  $X_0$ , in the sense that

$$E[(X_0 - E[X_0|\mathcal{G}_t])^2] \rightarrow 0 \quad \text{as } t \rightarrow \infty .$$

(Hint: Note that  $X_0 = e^{-rt}X_t$  and therefore  $E[X_0|\mathcal{G}_t] = e^{-rt}\hat{X}_t$ , so that

$$E[(X_0 - E[X_0|\mathcal{G}_t])^2] = e^{-2rt}S(t) .$$



- 6.4.** Consider the multi-dimensional linear filtering problem with no noise in the system:

$$\begin{aligned} \text{(system)} \quad & dX_t = F(t)X_t dt ; \\ & X_t \in \mathbf{R}^n, \quad F(t) \in \mathbf{R}^{n \times n} \end{aligned} \quad (6.3.12)$$

$$\begin{aligned} \text{(observations)} \quad & dZ_t = G(t)X_t dt + D(t)dV_t ; \\ & G(t) \in \mathbf{R}^{m \times n}, \quad D(t) \in \mathbf{R}^{m \times r} \end{aligned} \quad (6.3.13)$$

Assume that  $S(t)$  is nonsingular and define  $R(t) = S(t)^{-1}$ . Prove that  $R(t)$  satisfies the *Lyapunov equation* (compare with Exercise 6.2)

$$R'(t) = -R(t)F(t) - F(t)^T R(t) + G(t)^T (D(t)D(t)^T)^{-1} G(t). \quad (6.3.14)$$

(Hint: Note that since  $S(t)S^{-1}(t) = I$  we have  $S'(t)S^{-1}(t) + S(t)(S^{-1})'(t) = 0$ , which gives

$$(S^{-1})'(t) = -S^{-1}(t)S'(t)S^{-1}(t).)$$

- 6.5.** (Prediction)

In the *prediction problem* one seeks to estimate the value of the system  $X$  at a *future* time  $T$  based on the observations  $\mathcal{G}_t$  up to the present time  $t < T$ .

Prove that in the linear setup (6.2.3), (6.2.4) the predicted value

$$E[X_T | \mathcal{G}_t], \quad T > t$$

is given by

$$E[X_T | \mathcal{G}_t] = \exp \left( \int_t^T F(s) ds \right) \cdot \hat{X}_t. \quad (6.3.15)$$

(Hint: Use formula (6.2.16).)

- 6.6.** (Interpolation/smoothing)

The *interpolation* or *smoothing problem* consists of estimating the value of the system  $X$  at a time  $s < t$ , given the observations up to time  $t$ ,  $\mathcal{G}_t$ .

With notation as in (6.2.1), (6.2.2) one can show that  $M_s := E[X_s | \mathcal{G}_t]$  satisfies the differential equation

$$\begin{cases} \frac{d}{ds} M_s = F(s)M_s + C(s)C^T(s)S^{-1}(s)(M_s - \hat{X}_s); & s < t \\ M_t = \hat{X}_t. \end{cases} \quad (6.3.16)$$

(See Davis (1977, Theorem 4.4.4).)

Use this result to find  $E[X_s | \mathcal{G}_t]$  in Example 6.2.9.

**6.7.** Consider the system

$$dX_t = \begin{bmatrix} dX_1(t) \\ dX_2(t) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad E[X_0] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with observations

$$\begin{bmatrix} dZ_1(t) \\ dZ_2(t) \end{bmatrix} = \begin{bmatrix} X_1 \\ X_1 + X_2 \end{bmatrix} dt + \begin{bmatrix} dV_1(t) \\ dV_2(t) \end{bmatrix}.$$

Apply (6.3.14) from Exercise 6.4 to prove that  $S(t) := E[(X_t - \hat{X}_t)(X_t - \hat{X}_t)^T]$  is given by

$$S(t)^{-1} = S^{-1}(0) + \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} t$$

if  $S(0)$  is invertible. Then show that

$$d\hat{X}_t = -S(t) \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \hat{X}_t dt + S(t) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} dZ_t.$$

**6.8.** Transform the following Stratonovich equation

$$dX_t = b(t, X_t)dt + \sigma(t, X_t) \circ dB_t$$

into the corresponding Itô equation

$$dX_t = \tilde{b}(t, X_t)dt + \sigma(t, X_t)dB_t$$

using (6.1.3):

a)

$$\begin{bmatrix} dX_1 \\ dX_2 \end{bmatrix} = \begin{bmatrix} 1 \\ X_2 + e^{2X_1} \end{bmatrix} dt + \begin{bmatrix} 0 \\ e^{X_1} \end{bmatrix} \circ dB_t \quad (B_t \in \mathbf{R})$$

b)

$$\begin{bmatrix} dX_1 \\ dX_2 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} dt + \begin{bmatrix} X_2 \\ X_1 \end{bmatrix} \circ dB_t \quad (B_t \in \mathbf{R})$$

**6.9.** Transform the following Itô equation

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t$$

into the corresponding Stratonovich equation

$$dX_t = \hat{b}(t, X_t)dt + \sigma(t, X_t) \circ dB_t,$$

using (the converse of) (6.1.3):

a)  $dX_t = -\frac{1}{2}X_t dt + KX_t dB_t$ , where

$$K = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad X_t = \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix} \in \mathbf{R}^2 \quad \text{and} \quad B_t \in \mathbf{R}$$

(i.e.  $X_t$  is Brownian motion on the unit circle (Example 5.1.4)).

b) 
$$\begin{bmatrix} dX_1 \\ dX_2 \end{bmatrix} = \begin{bmatrix} X_1 & -X_2 \\ X_2 & X_1 \end{bmatrix} \begin{bmatrix} dB_1 \\ dB_2 \end{bmatrix}.$$

**6.10.** (On the support of an Itô diffusion)

The *support* of an Itô diffusion  $X$  in  $\mathbf{R}^n$  starting at  $x \in \mathbf{R}^n$  is the smallest closed set  $F$  with the property that

$$X_t(\omega) \in F \quad \text{for all } t \geq 0, \text{ for a.a. } \omega.$$

In Example 5.1.4 we found that Brownian motion on the unit circle,  $X_t$ , satisfies the (Itô) stochastic differential equation

$$\begin{bmatrix} dX_1(t) \\ dX_2(t) \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix} dt + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix} dB_t. \quad (6.3.17)$$

From this equation it is not at all apparent that its solution is situated on the same circle as the starting point. However, this can be detected by proceeding as follows: First transform the equation into its Stratonovich form, which in Exercise 6.9 is found to be

$$\begin{bmatrix} dX_1(t) \\ dX_2(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix} \circ dB_t. \quad (6.3.18)$$

Then (formally) replace  $\circ dB_t$  by  $\phi'(t)dt$ , where  $\phi$  is some smooth (deterministic) function,  $\phi(0) = 0$ . This gives the deterministic equation

$$\begin{bmatrix} dX_1^{(\phi)}(t) \\ dX_2^{(\phi)}(t) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \phi'(t)dt. \quad (6.3.19)$$

If  $(X_1^{(\phi)}(0), X_2^{(\phi)}(0)) = (1, 0)$  the solution of (6.3.19) is

$$\begin{bmatrix} X_1^{(\phi)}(t) \\ X_2^{(\phi)}(t) \end{bmatrix} = \begin{bmatrix} \cos \phi(t) \\ \sin \phi(t) \end{bmatrix}.$$

So for any smooth  $\phi$  the corresponding solution  $X^{(\phi)}(t)$  of (6.3.19) has its support on this unit circle. We can conclude that the original solution  $X(t, \omega)$  is supported on the unit circle also, in virtue of the *Stroock-Varadhan support theorem*. This theorem says that, quite generally, the support of an Itô diffusion  $X_t(\omega)$  coincides with the closure (in  $C([0, \infty); \mathbf{R}^n)$ ) of  $\{X^{(\phi)}(\cdot); \phi \text{ smooth}\}$ , where  $X^{(\phi)}(t)$  is obtained by replacing  $\circ dB_t$  by  $\phi'(t)dt$  in the same way as above. See e.g. Ikeda

and Watanabe (1989, Th. VI. 8.1). (In this special case above the support could also have been found directly from (6.3.18)).

Use the procedure above to find the support of the process  $X_t \in \mathbf{R}^2$  given by

$$dX_t = \frac{1}{2}X_t dt + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} X_t dB_t .$$

- 6.11.** Consider Example 6.2.1, but now without the assumption that  $E[X] = 0$ . Show that

$$\hat{X}_k = \frac{m^2}{ka^2 + m^2} E[X] + \frac{a^2}{a^2 + \frac{1}{k}m^2} \bar{Z}_k ; \quad k = 1, 2, \dots$$

(Compare with (6.2.8).)

(Hint: Put  $\xi = X - E[X]$ ,  $\zeta_k = Z_k - E[X]$ . Then apply (6.2.8) with  $X$  replaced by  $\xi$  and  $Z_k$  replaced by  $\zeta_k$ .)

- 6.12.** Prove formula (6.2.16).

(Hint:  $\exp \left( - \int_r^s F(u) du \right)$  is an integrating factor for the stochastic differential equation (6.2.3).)

- 6.13.** Consider the 1-dimensional linear filtering problem (6.2.3), (6.2.4). Find

$$E[\hat{X}_t] \quad \text{and} \quad E[(\hat{X}_t)^2] .$$

(Hint: Use Theorem 6.1.2 and use the definition of the mean square error  $S(t)$ .)

- 6.14.** Let  $B_t$  be 1-dimensional Brownian motion.

a) Give an example of a process  $Z_t$  of the form

$$dZ_t = u(t, \omega) dt + dB_t$$

such that  $Z_t$  is a Brownian motion w.r.t.  $P$  and  $u(t, \omega) \in \mathcal{V}$  is not identically 0.

(Hint: Choose  $Z_t$  to be the innovation process (6.2.13) in a linear filtering problem with  $D(t) \equiv 1$ .)

b) Show that the filtration  $\{\mathcal{Z}_t\}_{t \geq 0}$  generated by a process  $Z_t$  as in a) must be strictly smaller than  $\{\mathcal{F}_t\}_{t \geq 0}$ , i.e. show that

$$\mathcal{Z}_t \subseteq \mathcal{F}_t \quad \text{for all } t \text{ and } \mathcal{Z}_t \neq \mathcal{F}_t \text{ for some } t .$$

(Hint: Use Exercise 4.12.)